

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 State and explain parallelogram law of forces.

Q.24 State and explain parallel axis theorem.

Q.25 Derive with suitable notations of Torsion Equation.

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2nd Year / Advance Diploma in Tool and Die Making
Subject : Applied Mechanics and Strength of Materials

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Number of forces acting on body in equilibrium is:-

- a) One b) Two
- c) Three d) can be any number

Q.2 Frictional force always acts _____ to surface of application of the friction.

- a) Tangential b) Perpendicular
- c) Parallel d) Normal

Q.3 Forces whose line of action lie on same line known as:

- a) Coplanar forces b) Concurrent forces
- c) Collinear forces d) None

Q.4 Unit of force in SI units is

- a) Kg b) Newton
- c) Watt d) Dyne

Q.5 Possible loading in various members of framed structures are

- a) Compression b) Buckling
- c) Shear d) All

Q.6 Which is not the unit of pressure?

- a) Kg/cm b) Atmosphere
- c) Pascal d) Newton

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define units.

Q.8 Write one example of vector Quantities.

Q.9 What is Torsion?

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Q.10 Define Moments.

Q.11 A closed helical spring subjected to axial load, mainly develops _____ stress.

Q.12 Relation of springs in series. _____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain system of forces.

Q.14 Explain Newton's third law with example.

Q.15 Explain Lami's theorem.

Q.16 Write short note on hook's law.

Q.17 What do you understand by Poisson's ratio?

Q.18 Define principle of Torque.

Q.19 Define Torsional Rigidity.

Q.20 Explain stiffness.

Q.21 Define deflection in Leaf springs.

Q.22 Explain Strain energy.

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